Alberto del Río Jara

Fayetteville, AR | 479-480-0922 | albertod@uark.edu | Portfolio | LinkedIn | GitHub

M.S. Statistics and Analytics candidate (GPA 4.0, May 2027) with applied experience in business intelligence, cloud data pipelines, analytics engineering, and enterprise AI prototypes. Builds data solutions with Python, SQL, Power BI, Microsoft Fabric, BigQuery, Airflow, dbt, and Terraform.

TECHNICAL SKILLS

Programming & Analytics: Python, SQL, R, Pandas, NumPy, scikit-learn, statistical analysis, predictive modeling
Data Engineering & Cloud: BigQuery, Google Cloud Storage, Airflow, dbt, Terraform, Docker, ETL/ELT, data quality testing
Business Intelligence: Power BI, Microsoft Fabric, Looker Studio, Tableau, Excel, dashboard development
Development: Git, GitHub Actions, CI/CD

EDUCATION

M.S. in Statistics and Analytics — Computational Analytics Expected May 2027
University of Arkansas, Fayetteville, AR | GPA: 4.0/4.0

B.S. in Telecommunication Technologies Engineering (Electrical and Computer Engineering curriculum) May 2025
Carlos III University of Madrid, Spain | Bilingual Program | Exchange Student, University of Arkansas, 2024–2025

EXPERIENCE

AI Innovation Analyst Intern Aug 2026 – Present
University of Arkansas
- Prototype an HR AI solution in Microsoft Fabric IQ using ontologies, semantic models, and data agents to support retrieval from enterprise HR data.
- Design and evaluate prompts and retrieval workflows; define performance metrics to assess response quality and prototype behavior.
- Document findings and implementation considerations for applying semantic and agent-based AI capabilities to university HR use cases.

Graduate Assistant — Strategic Data and Process Innovation Aug 2025 – Present
University of Arkansas
- Delivered Power BI reports and analytics solutions for HCM, Data Strategy, FAMA, and other university units, enabling data-informed operational decisions through SQL-based modeling and Microsoft Fabric.
- Improved the availability and reliability of institutional data by contributing to data-engineering pipelines supporting reporting and analytics across multiple university departments.
- Advanced the UARK AI Administration initiative with the Computer Vision and Image Understanding Lab by contributing to the development and evaluation of AI solutions for university administrative processes.

Embedded Software Engineer Intern Jul 2024
JARMAR
- Contributed to embedded C software for monitoring solar-panel performance using Silicon Labs tools and ESP32 technology.

Academic Tutor 2019 – 2022
Freelance
- Tutored high school students in mathematics, physics, and English for up to 10 hours per week while completing a full-time degree.

SELECTED PROJECTS

End-to-End Cloud Data Pipeline for Housing Analytics | GitHub Summer 2026
- Built a 13-task Airflow DAG to ingest housing data from CSV into Google Cloud Storage and BigQuery.
- Developed 10 dbt models with 61 passing tests and automated cloud infrastructure and CI with Terraform and GitHub Actions.
- Created a Looker Studio dashboard for occupancy, bed availability, and student assignments by building.

ML Classifier for Fault Diagnosis in Rotary Machines | GitHub Spring 2025
- Built and compared SVM, Random Forest, and other classifiers to detect early-stage component faults from vibration and performance data.
- Achieved 96-98% multiclass accuracy across 38 rotary-machine fault categories by training Decision Tree models on engineered time-domain vibration features.
- Benchmarked 60 model and data configurations across 5 machine-learning algorithms, 3 operating speeds, and time- and frequency-domain features using accuracy, precision, recall, F1-score, and confusion matrices.
- Processed 2,850 vibration-signal CSV files covering 38 fault scenarios by building Python feature-engineering pipelines with Pandas, NumPy, SciPy, FFT, and statistical signal features.

LANGUAGES

Spanish: Native | **English:** Fluent (C1)